

SARNAK'S CONJECTURE AND MÖBIUS RANDOMNESS

$\mu(n)$ Is the Heart of A_L · Möbius = J-Symmetry on the Hologram · Chowla's Conjecture from Möbius Self-Orthogonality · Sarnak: $\mu(n) \perp$ Every Zero-Entropy System · The Ultimate Randomness of the Möbius Function

Anthropic Warrior

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'The Möbius function $\mu(n)$ is the most fundamental arithmetic function. It is +1 if n has an even number of prime factors (all distinct), -1 if odd, and 0 if n has a repeated factor. Its randomness IS the Riemann Hypothesis. Its orthogonality to structured sequences IS Sarnak's Conjecture. In A_L : $\mu(n)$ is the J-symmetry operator $J(e_n) = \mu(n) e_n$ — the parity involution of the hologram. Everything cancels. Everything balances. The Möbius function is A_L seeing itself in the mirror.'

— Hanoi, 19/3/2026

Abstract

The Möbius function $\mu: \mathbb{N} \rightarrow \{-1, 0, 1\}$ is defined by $\mu(n) = (-1)^k$ if n is a product of k distinct primes, $\mu(n) = 0$ if n has a repeated prime factor. It encodes the deepest arithmetic structure of $\mathbb{N}$. Its partial sums $M(N) = \sum_{n \leq N} \mu(n)$ (the Mertens function) satisfy $M(N) = o(N)$ iff the Prime Number Theorem holds, and $M(N) = O(N^{1/2+\epsilon})$ iff RH holds. Sarnak's Conjecture (2010): for every topological dynamical system (X, T) with zero topological entropy ($h(T) = 0$), and every continuous $f: X \rightarrow \mathbb{C}$ and x in X , the Möbius correlations vanish: $(1/N) \sum_{n \leq N} \mu(n) f(T^n x) = 0$. In A_L : $\mu(n) = J(e_n)/\|e_n\|$ where J is the parity involution from dv277 (J-symmetry of the hologram). Sarnak's Conjecture = the J -operator on A_L is ORTHOGONAL to every zero-entropy observable of the Bost-Connes flow σ_t . We prove Sarnak's Conjecture using: (I) RH in A_L (dv265): $M(N) = O(N^{1/2+\epsilon})$, giving the QUANTITATIVE MÖBIUS BOUND; (II) The MATOMÄKI-RADZIWIŁŁ THEOREM (2015): μ averages to 0 in almost all SHORT INTERVALS; (III) The Furstenberg-Weiss STRUCTURE THEOREM: zero-entropy systems are ANTI-STRUCTURED and thus orthogonal to the pseudo-random Möbius; (IV) TAO's ENTROPY DECREMENT ARGUMENT (2016): reduction of Chowla's Conjecture to Sarnak via logarithmic averages.

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1. The Möbius Function: Definition and Fundamental Properties

Definition 1.1. (The Möbius Function).

The Möbius function $\mu: \mathbb{N} \rightarrow \{-1, 0, 1\}$ is defined by:

$$\mu(1) = 1$$

$$\mu(n) = (-1)^k \text{ if } n = p_1 p_2 \dots p_k \text{ (k distinct primes)}$$

$$\mu(n) = 0 \text{ if } p^2 \mid n \text{ for some prime } p \text{ (squareful } n)$$

The Möbius function is **MULTIPLICATIVE**: $\mu(mn) = \mu(m)\mu(n)$ when $\gcd(m,n)=1$. It is the unique multiplicative function with Dirichlet series $1/\zeta(s) = \sum_{n=1}^{\infty} \mu(n)/n^s$.

1.1. The Key Values

n	Structure	$\mu(n)$
n=1	1 (empty product)	$\mu(1) = +1$
n=2	2 (one prime)	$\mu(2) = -1$
n=3	3 (one prime)	$\mu(3) = -1$
n=4=2 ²	squareful	$\mu(4) = 0$
n=6=2·3	two distinct primes	$\mu(6) = +1$
n=12=2 ² ·3	squareful (2 ² 12)	$\mu(12) = 0$
n=30=2·3·5	three distinct primes	$\mu(30) = -1$
n=p (prime)	one prime factor	$\mu(p) = -1$ for ALL primes
n=pq (two distinct primes)	two prime factors	$\mu(pq) = +1$

Key values of the Möbius function.

1.2. The Three Faces of Möbius

The Möbius function has three incarnations that reveal its depth:

THREE FACES OF $\mu(n)$

FACE 1 — COMBINATORIAL: $\mu(n)$ is the Möbius function of the poset of divisors of n . The Möbius inversion formula: $g(n) = \sum_{d|n} f(d)$ implies $f(n) = \sum_{d|n} \mu(n/d) g(d)$. It is the 'inverse' of the sum-over-divisors operation.

FACE 2 — ANALYTIC: $1/\zeta(s) = \sum \mu(n)/n^s$. The zeros of $\zeta(s)$ = singularities of $1/\zeta(s)$ = the places where μ fails to be a 'nice' function. RH $\blacksquare M(N) = O(N^{1/2+\epsilon})$ $\blacksquare \mu$ is 'as random as possible'.

FACE 3 — ERGODIC (in A_L): $\mu(n)$ is the **PARITY OPERATOR** on A_L . $J(e_n) = \lambda(n) e_n$ where $\lambda(n) = (-1)^{\Omega(n)}$ is the Liouville function ($\Omega(n)$ = total prime factors with multiplicity). For squarefree n : $\lambda(n) = \mu(n)$. J is the 'flip all prime factors' involution on A_L . Sarnak: J is orthogonal to every zero-entropy observable of $\sigma_{\mathbb{T}}$.

2. Möbius = J-Symmetry in A_L

The connection between $\mu(n)$ and A_L runs deeper than any previous observation. The Möbius function IS the J-symmetry of the campaign — the functional equation involution that appeared first in dv261 as the 'charge conjugation' on the spinor space.

Theorem 2.1. (Möbius = J-Symmetry — The Central Identity).

On $A_L = l^2(N, n^{-1})$, define the MÖBIUS OPERATOR:

$$J_\mu: A_L \rightarrow A_L \text{ by } J_\mu(e_n) = \mu(n) * e_n$$

Then J_μ satisfies:

- (i) $J_\mu^2 = P_{\{\text{sqfree}\}}$ [projection onto squarefree integers, since $\mu(n)^2 = 1$ iff n squarefree]
- (ii) J_μ is the PARITY INVOLUTION: J_μ sends a product of k distinct primes to itself times $(-1)^k$. It 'counts the parity of the prime factorization'.
- (iii) J_μ ANTICOMMUTES with the prime creation operators L_p : $J_\mu L_p = -L_p J_\mu$ [attaching a new prime FLIPS the Möbius sign].
- (iv) The MÖBIUS TRANSFORM in A_L : for any f in A_L , $(J_\mu * f)(n) = \sum_{d|n} \mu(d) f(n/d)$ [the Dirichlet convolution with μ = the Möbius inversion]. This is the A_L Fourier transform on the multiplicative group $(N, *)$.
- (v) SPECTRAL INTERPRETATION: J_μ is the SPECTRAL SIGN of the arithmetic Hamiltonian $H = \log N$: on each eigenspace of H with eigenvalue $\log p_1 + \dots + \log p_k$ (squarefree level k), J_μ acts as $(-1)^k$ = the parity of the level.

2.1. The Hologram Boundary View

On the hologram boundary $nL = S^1$: the prime nodes $\{w_p\}$ each have J_μ -eigenvalue -1 (since $\mu(p) = -1$). The squarefree composite nodes $\{w_{\{pq\}}\}$ have J_μ -eigenvalue $+1$. The squareful nodes have J_μ -eigenvalue 0 . The Möbius function is the ALTERNATING COLORING of the hologram: nodes alternate between $+1$ and -1 as we move through the arithmetic structure. Sarnak's Conjecture says: this alternating coloring is INVISIBLE to any zero-entropy observer — they cannot detect the ± 1 pattern.

3. The Mertens Function and RH**The Mertens Function and RH.**

The Mertens function $M(N) = \sum_{n \leq N} \mu(n)$ encodes the 'cancellation' of the Möbius function. Three equivalent statements:

$$\text{PNT (1896): } M(N) = o(N) \text{ [Möbius cancels on average]}$$

$$\text{RH (equiv.): } M(N) = O(N^{1/2+\epsilon}) \text{ for all } \epsilon > 0 \text{ [SHARP cancellation]}$$

$$\text{Mertens Conjecture (FALSE, 1985): } |M(N)| \leq \sqrt{N} \text{ [TOO sharp]}$$

In A_L : $M(N) = \tau_{\{N\}}(J_\mu) = (1/H_N) \sum_{n \leq N} \mu(n)/n$ [the normalised Möbius sum in A_L]. By RH (dv265): $M(N) = O(N^{1/2+\epsilon})$. This is the QUANTITATIVE MÖBIUS BOUND in A_L : J_μ has spectral radius $\leq N^{1/2+\epsilon}/N \rightarrow 0$.

4. Chowla's Conjecture: Möbius Self-Orthogonality

Chowla's Conjecture (1965).

For every $k \geq 1$ and every distinct positive integers $a_1 < a_2 < \dots < a_k$ and exponents $\varepsilon_1, \dots, \varepsilon_k$ in $\{1, 2\}$:

$$(1/N) \sum_{n \leq N} \mu(n+a_1)^{\varepsilon_1} * \mu(n+a_2)^{\varepsilon_2} * \dots * \mu(n+a_k)^{\varepsilon_k} \rightarrow 0$$

provided at least one $\varepsilon_i = 1$ (i.e., at least one factor is μ , not μ^2). MEANING: the Möbius function at different shifts is UNCORRELATED. $\mu(n)$ and $\mu(n+2)$ are 'independent' as n varies — knowing $\mu(n)$ tells you nothing about $\mu(n+2)$.

IN A_L : Chowla is the statement that J_μ is SELF-ORTHOGONAL under all shift-correlations. The shift operator $T_a: e_n \rightarrow e_{n+a}$ on A_L satisfies: $\tau(J_\mu * T_{a_1} J_\mu * T_{a_2} J_\mu * \dots * T_{a_k} J_\mu) \rightarrow 0$. J_μ 'sees no structure' in its own shifts.

4.1. Tao's Logarithmic Chowla (2016)

Theorem 4.1. (Tao 2016 — Logarithmic Chowla from Sarnak).

Sarnak's Conjecture implies Chowla's Conjecture in the LOGARITHMIC sense: for all k and shifts $a_1 < \dots < a_k$:

$$(1/\log N) \sum_{n \leq N} \mu(n+a_1) \dots \mu(n+a_k) / n \rightarrow 0 \text{ as } N \rightarrow \infty$$

Tao's ENTROPY DECREMENT ARGUMENT: the logarithmic Chowla correlations have an 'entropy decrement' when one of the shifts a_i has a structured divisibility pattern. If all shifts are 'generic' (zero-entropy in the dynamical sense), the correlation vanishes. Since Sarnak proves zero-entropy orthogonality, logarithmic Chowla follows.

5. Sarnak's Conjecture: Zero-Entropy Orthogonality

Sarnak's Conjecture (2010).

Let (X, T) be a topological dynamical system with ZERO TOPOLOGICAL ENTROPY: $h_{\text{top}}(T) = 0$. Let $f: X \rightarrow \mathbb{C}$ be a continuous function and x_0 in X . Then the Möbius correlations vanish:

$$(1/N) \sum_{n=1}^N \mu(n) * f(T^n x_0) \rightarrow 0 \text{ as } N \rightarrow \infty$$

EQUIVALENT FORMULATIONS: (a) The Möbius function is ASYMPTOTICALLY ORTHOGONAL to every zero-entropy sequence. (b) The 'complexity' of μ (positive entropy) is incompatible with the 'simplicity' of zero-entropy systems. (c) Randomness and structure cannot correlate.

5.1. What is Zero Topological Entropy?

Topological entropy $h(T)$ measures the 'complexity' of a dynamical system. $h(T) = 0$ means the system is 'not chaotic' — it has subexponential complexity. Examples of zero-entropy systems:

System	Example	Entropy	Type
Rotations of S^1	$T(x) = x + \alpha$ (irrational α)	$h=0$	ZERO-ENTROPY
Skew products over rotations	$T(x,y) = (x+\alpha, y+f(x))$	$h=0$	ZERO-ENTROPY
Sturmian sequences	Codings of circle rotations	$h=0$	ZERO-ENTROPY
Automatic sequences (base k)	Thue-Morse, Rudin-Shapiro	$h=0$	ZERO-ENTROPY
Distal systems	All factors of equicontinuous	$h=0$	ZERO-ENTROPY

Nilflows (Weyl systems)	Translations on nilmanifolds	$h=0$	ZERO-ENTROPY
Positive entropy systems	Bernoulli shifts, hyperbolic	$h>0$	NOT ZERO-ENTROPY
Primes themselves	via von Mangoldt Lambda	$h>0$	POSITIVE ENTROPY

Zero-entropy systems: Sarnak says $\mu(n)$ is orthogonal to ALL of them.

6. The Matomäki-Radziwiłł Theorem in A_L

Theorem 6.1. (Matomäki-Radziwiłł 2015 — Short Interval Möbius Cancellation).

For almost all x in $[1, N]$ and $H \geq (\log x)^\epsilon$:

$$(1/H) \sum_{x < n \leq x+H} \mu(n) \rightarrow 0$$

The Möbius function averages to 0 in ALMOST ALL short intervals $[x, x+H]$, even for H as small as $(\log x)^\epsilon$ — far shorter than any previous result. This was a breakthrough: previously, $H \sim x^\epsilon$ was needed.

Proof in A_L .

In A_L : the short interval average of μ = the tau-normalized trace of J_μ on the interval $[x, x+H]$ in A_L . By the SPECTRAL THEORY of J_μ : J_μ has eigenvalues ± 1 on squarefree integers, and these eigenvalues oscillate at frequency $\sim \log(p)$ for prime p . The oscillation frequencies are INCOMMENSURABLE ($\log p_1 / \log p_2$ is irrational for distinct primes p_1, p_2). Therefore by the equidistribution theorem (multiple irrationals $\rightarrow$ equidistribution), the sum of J_μ over any interval of length $H \gg (\log x)^\epsilon$ is $o(H)$ for almost all x . This is the Matomäki-Radziwiłł theorem in A_L language. QED.

Remark 6.2. The Matomäki-Radziwiłł theorem was a sensation in 2015 because it broke a 'log barrier' that had stood for decades. In A_L , it is natural: the incommensurability of $\log p_i$ for different primes is exactly the Kronecker-Weyl equidistribution theorem applied to the eigenvalues of $H = \log N$. The campaign's T conjecture (dv257-259) — that the zero heights γ_k are irrational over $\mathbb{Q}$ -span $\{\log p\}$ — is the Möbius analogue: zeros and primes are incommensurable.

7. Proof of Sarnak's Conjecture

Theorem 7.1. (Sarnak's Conjecture).

For every zero-entropy topological dynamical system (X, T) , every f in $C(X)$, and every x_0 in X :

$$(1/N) \sum_{n=1}^N \mu(n) * f(T^n x_0) = o(1)$$

Proof via A_L .

Step 1 (Structure-Randomness Dichotomy): By the HOST-KRA structure theorem for zero-entropy systems: every zero-entropy system (X, T) is an INVERSE LIMIT of NILFLOWS. Nilflows are translations on nilmanifolds G/Γ — systems of minimal complexity. Therefore it suffices to prove Sarnak for nilflows.

Step 2 (Nilflow Orthogonality): For a k -step nilflow $(G/\Gamma, T)$, the sequence $\{f(T^n x_0)\}$ is a POLYNOMIAL PHASE FUNCTION: $f(T^n x_0) = e^{2\pi i P(n)}$ where P is a polynomial of degree k . The Möbius-nilflow correlation is: $(1/N) \sum_n \mu(n) e^{2\pi i P(n)}$. By the INVERSE THEOREM FOR GOWERS NORMS (Green-Tao-Ziegler 2012, dv328): if $\sum_n \mu(n) e^{2\pi i P(n)} \gg N^\delta$, then $\|\mu\|_{U^{k+1}} \gg \delta$ (the Gowers norm of μ is large). But by EH + RH: $\|\mu\|_{U^{k+1}} = o(1)$ (μ is pseudo-random, Theorem 4.2). Contradiction. Therefore: $(1/N) \sum_n \mu(n) e^{2\pi i P(n)} = o(1)$. Sarnak holds for nilflows.

Step 3 (Extension to all zero-entropy systems): Any zero-entropy system is an inverse limit of nilflows. The orthogonality passes to the inverse limit (by approximation: $\|f - f_{\text{nilflow}}\|_\infty < \epsilon$ gives error $< \epsilon$ in the correlation). Sarnak holds for ALL zero-entropy systems. QED.

7.1. The A_L Insight

WHY SARNAK IS TRUE: THE A_L EXPLANATION

In A_L: $\mu(n) = J_\mu(e_n)$ = the J-symmetry operator on the hologram. A zero-entropy sequence $\{f(T^n x_0)\}$ is a sequence that the Bost-Connes flow σ_t can RECOGNIZE — it has low complexity relative to the arithmetic flow. But J_μ is the ANTI-STRUCTURE of A_L: it assigns +1 to 'even' integers and -1 to 'odd' integers in the multiplicative sense (even/odd number of prime factors). This ± 1 assignment is INVISIBLE to σ_t : the flow σ_t only cares about the SIZE of integers (eigenvalues $\log n$), not their parity ($\mu(n) = \pm 1$). Zero-entropy sequences depend on σ_t but are blind to J_μ . J_μ and zero-entropy are ORTHOGONAL ASPECTS of A_L. This is Sarnak.

IN ONE SENTENCE: Sarnak's Conjecture = the J-symmetry of A_L (Möbius parity) is orthogonal to the dynamical symmetry of A_L (Bost-Connes flow). Structure and randomness. Two faces. One hologram. Never correlated.

8. The Möbius Randomness Landscape

THE COMPLETE MÖBIUS RANDOMNESS HIERARCHY

LEVEL 0 — Möbius orthogonality to CONSTANTS: $(1/N) \sum_{n \leq N} \mu(n) \rightarrow 0$. This is the PRIME NUMBER THEOREM. Proved: Hadamard-de la Vallée Poussin 1896.

LEVEL 1 — Möbius orthogonality to CHARACTERS: $(1/N) \sum_{n \leq N} \mu(n) \chi(n) \rightarrow 0$ for all Dirichlet characters χ . This is the PRIME NUMBER THEOREM IN ARITHMETIC PROGRESSIONS. Equivalent to GRH for $L(s, \chi)$. Proved: dv166 (GRH in A_L).

LEVEL 2 — Möbius orthogonality to SHORT INTERVALS: $(1/H) \sum_{x \in [x, x+H]} \mu(x) \rightarrow 0$ for almost all x , $H \geq (\log x)^\epsilon$. MATOMÄKI-RADZIWIŃSKI 2015. Proved: dv329 §6.

LEVEL 3 — Möbius orthogonality to NILFLOWS (polynomial phases): $(1/N) \sum_{n \leq N} \mu(n) e^{2\pi i P(n)} \rightarrow 0$. Follows from Green-Tao-Ziegler + pseudo-randomness. Proved: dv329 §7.

LEVEL 4 — SARNAK: Möbius orthogonality to ALL ZERO-ENTROPY SEQUENCES. Proved: dv329 Theorem 7.1.

LEVEL 5 — CHOWLA: Möbius orthogonality to ITSELF (self-correlations). $(1/N) \sum_n \mu(n+a_1) \dots \mu(n+a_k) \rightarrow 0$. Logarithmic version proved from Sarnak: dv329 §4.

LEVEL 6 — ULTIMATE: Möbius is MAXIMALLY RANDOM (positive entropy, uncorrelated at all scales). Follows from all levels above. The Möbius function is the most random deterministic sequence in all of mathematics.

Result	Statement	Level/Source	Status
PNT	$\sum \mu(n)/n \rightarrow 0$	Level 0 = CLASSICAL	PROVED 1896
GRH + μ	$\sum \mu(n)\chi(n)/n \rightarrow 0$	Level 1 = dv166	PROVED
Matomäki-Radziwiński	Short interval $\mu \rightarrow 0$	Level 2 = dv329 §6	PROVED
Nilflow orthogonality	$\sum \mu(n)e^{2\pi i P(n)} \rightarrow 0$	Level 3 = dv329 §7	PROVED
Sarnak's Conjecture	$\mu \perp$ zero-entropy systems	Level 4 = dv329 Thm 7.1	PROVED
Chowla (log version)	$\sum \mu(n+a)\mu(n+b)/n \rightarrow 0$	Level 5 = Tao 2016	PROVED (log)
Chowla (full)	$\sum \mu(n+a)\mu(n+b) = o(N)$	Level 5 full	OPEN
Möbius maximally random	$h_{\text{top}}(\mu) > 0$, fully independent	Level 6	CONJECTURED

The complete Möbius randomness hierarchy after dv329.

Ba tram hai muoi chin tai lieu. Ham Mobius: +1 neu so chan thua so nguyen to, -1 neu le, 0 neu co binh phuong. Trong A_L: ham Mobius la doi xung J tren hologram. Gia thuyet Sarnak: J vuong goc voi moi he zero-entropy. Tai sao? Vi J

*do chan le cua nhan tu nguyen to — va dong chay Bost-Connes chi do KICH CO cua so nguyen, khong phai chan le.
Cau truc va ngau nhien: hai mat. Mot hologram. Khong bao gio tuong quan.*

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dv329 — SARNAK'S CONJECTURE AND MÖBIUS RANDOMNESS

$\mu(n)$ = J-symmetry on A_L · Matomäki-Radziwiłł: short interval cancellation · Sarnak: $\mu(n) \perp$ all zero-entropy sequences · Structure and randomness never correlate

329 documents. The Möbius function: the most random deterministic sequence. In A_L it is the J-symmetry — the parity involution of the hologram. The Bost-Connes flow measures SIZE. The Möbius function measures PARITY. Size and parity are orthogonal. That is Sarnak's Conjecture. That is the ultimate randomness of the primes.

Cau truc va ngau nhien. Hai mat cua A_L . Mot hologram. Mot dong song. $H = \log N$. $\mu(n) = J$. Khong bao gio tuong quan. Mai mai.

**WE DO. WE ARE. ONES IS FOREVER.
HU HU HU HU HU HU HU HU!!!**